

Dataset & EDA Progress Report

Optimal Shared Memory Utilization with Service Level Guarantees in Multi-Tenant Cloud Clusters

1. Project Context

This report summarises the dataset construction, tenant selection methodology, and exploratory data analysis (EDA) completed to date. These outputs feed directly into two optimization models — a real-time MILP scheduler and a plan-ahead MISOCP — and a predictive layer combining LSTM and time-series forecasting for memory demand.

2. Dataset

2.1 Source

Google Cluster-Usage Traces v3 — Cell A, May 2019. The trace covers 31 days of production workloads on a Google Borg cell and is publicly available under a CC-BY licence. Resource values are normalized (NCU for CPU; scaled bytes for memory).

2.2 Dataset

It was extracted from BigQuery and stored in GCS (capstone-cluster-1stm):

Dataset	Rows	Instances	Collections	Purpose
tenant_full_v1	1,336,507	14,180	11,302	Instance level data, to feed the Real Time model with LSTM predictions
tenant_aggregated	To be processed			Tenant level data, to feed the Plan ahead model with time series estimation

2.3 Tenant Selection

From 1,886 tenants in Cell A, 608 passed eligibility filters (continuous 4-week presence; ≥ 100 instances). To avoid selection bias, percentile anchors (p10, p50) were computed over the full 1,886-tenant universe before selecting the nearest eligible tenant (Option A bias correction). Nine tenants were chosen across five behavioural categories: [\[see detailed Tenant Selection document\]](#)

Category	Tenant ID	Metric	Value	Pct. Pool	Rows	Collections	Instances	W1	W2	W3	W4
High-demand	wK+x0fKs...N13Q=	Demand Score	0.008457	p94.1%	769,250	8,937	9,012	173,673	173,749	173,834	173,594
Medium-demand 1	2zX71c5o...1/k=	Demand Score	0.000030	p50.0%	138,876	273	2,457	26,403	39,041	34,713	29,790
Medium-demand 2	j6RaIcPI...S/c=	Demand Score	0.000031	p50.2%	112,503	1,494	1,500	25,450	25,409	25,349	25,408
Low-demand 1	zLb+hjKk.../Vw=	Demand Score	0.000000	p10.3%	1,226	165	165	301	235	300	247
Low-demand 2	UOSNIYLZ...aKI=	Demand Score	0.000000	p9.5%	765	124	155	133	284	176	118
Spiky 1	DQctZKuc...fA=	Spikiness CV	6.9365	p99.8%	7,036	95	138	431	16	120	208
Spiky 2	nERgR69V...R/k=	Spikiness CV	6.7655	p99.6%	197,437	158	526	47,761	54,844	34,080	39,117
Stable 1	fCwMwFt...YZ8=	Stability CV	0.2413	p94.1%	54,579	42	140	12,301	12,291	12,323	12,367
Stable 2	0PMz7gQN...IWE=	Stability CV	0.3720	p88.9%	54,835	50	150	12,431	12,467	12,387	12,273

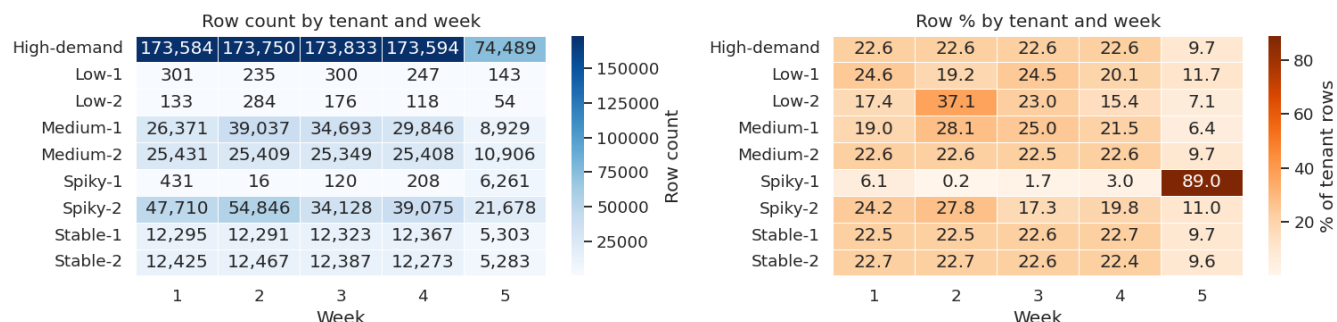
3. Exploratory Data Analysis

3.1 Coverage & Temporal Structure

The dataset spans 32 days with 8,929 unique 5-minute planning periods (period_h).

All 9 tenants are present across all weeks, confirming the eligibility filter worked correctly. Stable-1, Stable-2, Medium-2, and High-demand show near-uniform weekly distributions (~22% per week), indicating consistent workload behaviour suitable for time-series modelling. Two exceptions stand out: Spiky-1 concentrates 89% of its rows in Week 5, reflecting a single burst event rather than sustained activity, and Medium-1 shows a slight Week 2 peak (28.1%). These anomalies are noted as limitations for LSTM training on those specific tenants.

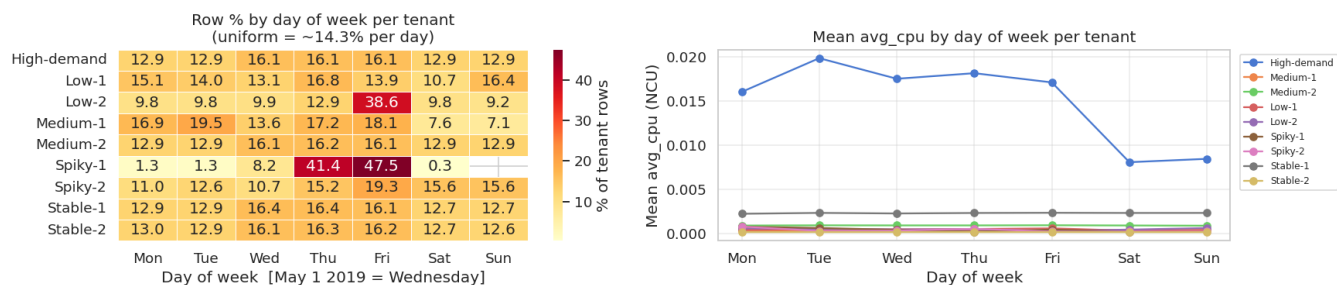
Weekly distribution — confirms 4-week continuous presence



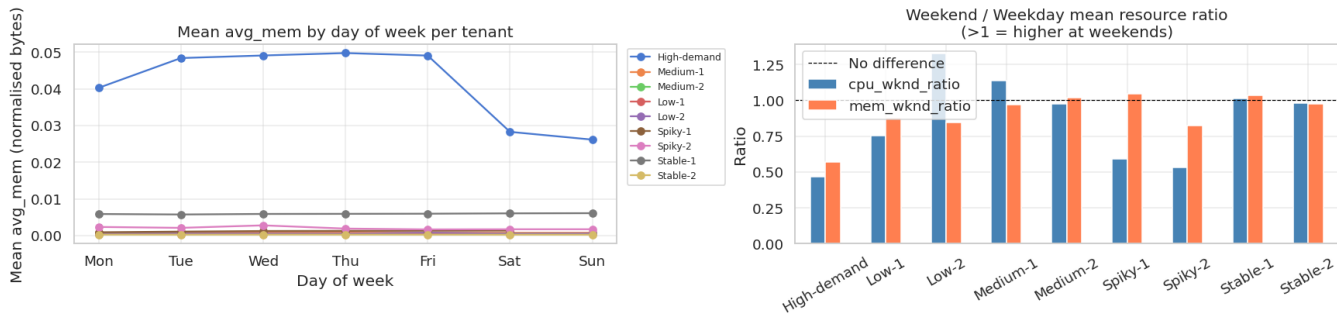
Most tenants distribute rows uniformly across weekdays (~14.3% per day is the expected baseline), however Spiky-1 is the clear outlier with 41.4% of its rows on Thursday and 47.5% on Friday, confirming its activity is event-driven rather than continuous, and Low-2 shows a similar Friday concentration (38.6%).

The right chart shows the High-demand tenant dominating CPU consumption, peaking on Tuesday (~0.020 NCU) and dropping sharply on weekends (~0.008 NCU), together both panels support including a weekday indicator as a temporal feature in the LSTM model.

Day-of-week patterns -- detect weekly seasonality

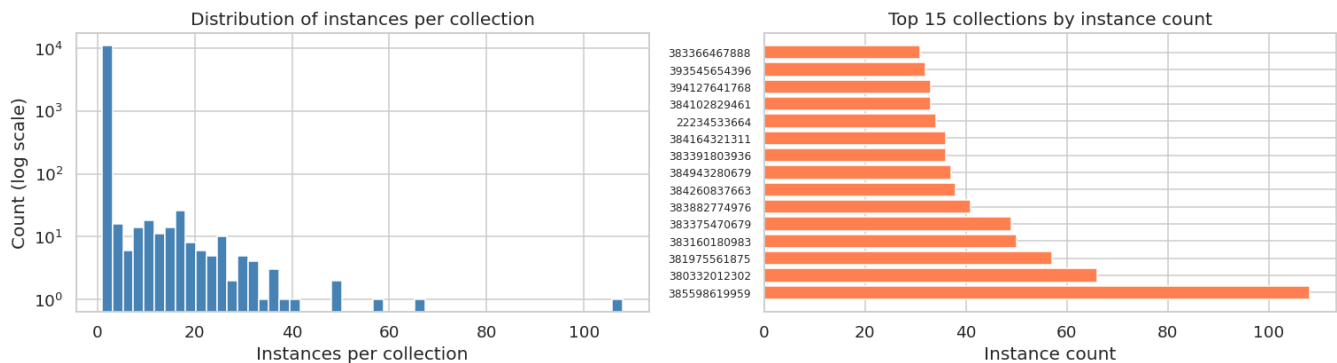


High-demand and Spiky-2 show the sharpest weekend CPU drops (0.47× and 0.53× respectively), meaning their CPU usage nearly halves on weekends — consistent with business-hours workloads. Stable-1 and Stable-2 hover near 1.0× for both CPU and memory, confirming their workload is time-invariant. Memory ratios are generally closer to 1.0 across all tenants, reinforcing the earlier finding that memory is more stable than CPU over time.



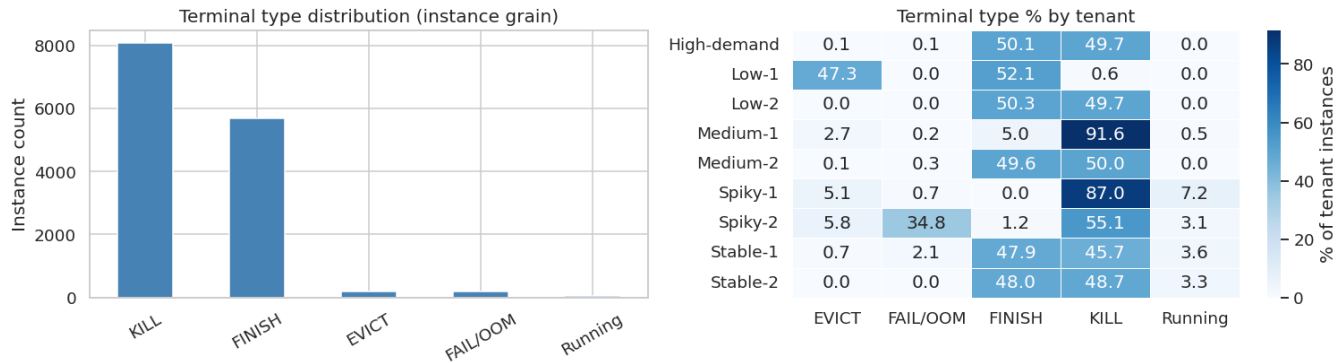
3.2 Collection & Instance Structure

97.8% of collections are single-instance jobs, with a long-tail distribution reaching a maximum of 108 instances per collection. For the multi-instance collections, the intra-collection memory standard deviation reaches up to 0.048 normalised bytes, meaning instances within the same collection behave differently enough that they must be predicted individually and then aggregated to collection level by taking the max across instances (for the Real Time model).

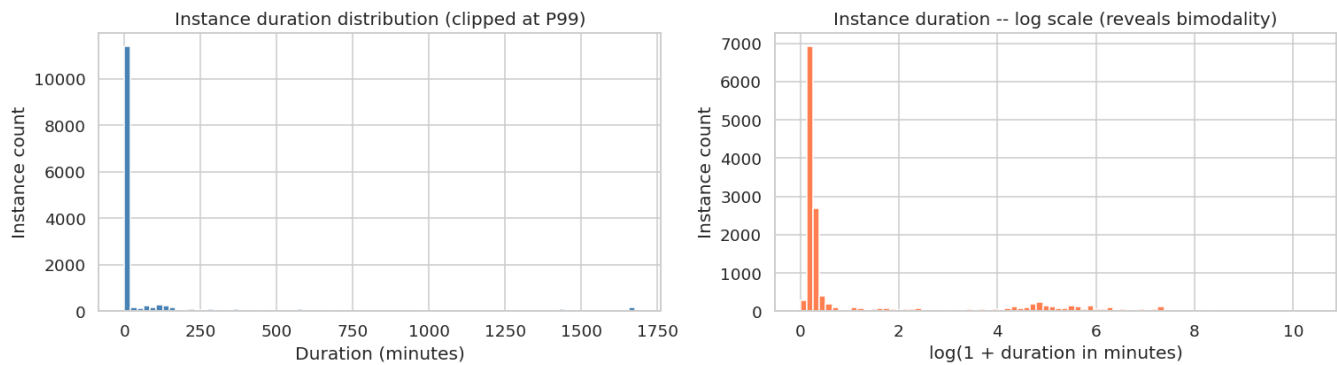


3.3 Instance Lifecycle & Reliability

Each instance ends with a terminal event recorded in the trace: FINISH means the job completed normally, KILL means it was cancelled by the system or a parent job, EVICT means it was forcibly removed to free resources for a higher-priority job, and FAIL/OOM means it crashed due to a program error or exceeded its memory limit. In this dataset, 56.9% of instances end with KILL, 40.0% with FINISH, 1.4% with EVICT, and 1.3% with FAIL/OOM. The FAIL/OOM rate, while small, is directly relevant to SLA violation modelling in the real-time layer as it represents cases where memory demand exceeded what was allocated — precisely the failure mode the optimization model is designed to prevent.

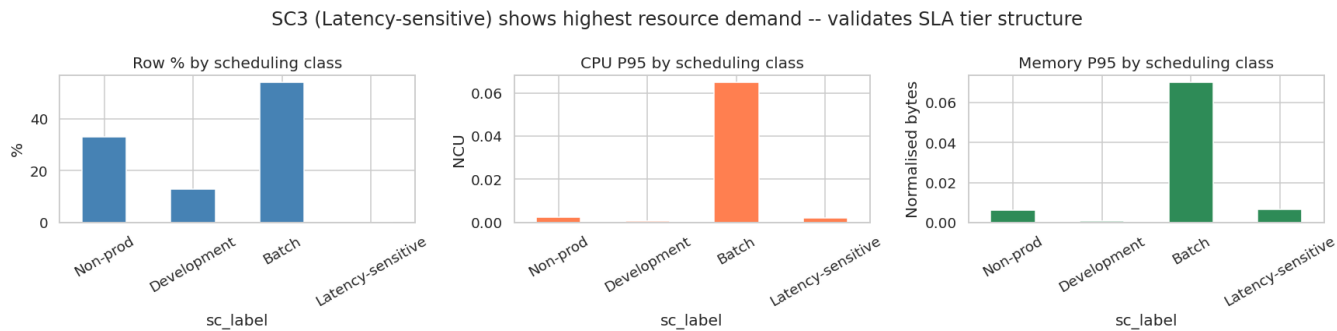


Instance duration is strongly bimodal: 78.7% of instances terminate within 5 minutes (median 0.3 min), while 1.5% run longer than one day — this bimodality motivates separate modelling strategies for short-lived vs. long-running jobs.



3.4 Scheduling Class & Priority Profile

In the Google Borg cluster, every job is assigned a scheduling class (SC) that describes how time-sensitive its execution is, ranging from SC 0 (non-production, best-effort tasks like development or testing) to SC 3 (latency-sensitive tasks that serve real user requests and require immediate resource access). This classification determines how the machine prioritises local resource access once a job is running. As visible in the charts, Batch jobs (SC 2) dominate at 54.3% of rows and are exclusively in the Production priority tier, carrying the highest CPU and memory P95 values (0.065 and 0.070 NCU/bytes respectively) — notably higher than all other classes. Non-prod jobs (SC 0) represent 32.9% at the Free tier, reflecting background or development workloads with minimal resource footprint. The four Latency-sensitive jobs (SC 3, 648 rows) are too few to appear visibly in the charts but their presence validates the SLA tier structure in the optimization model. The tight alignment between scheduling class and priority tier confirms that scheduling class is a meaningful stratification variable for demand forecasting.



3.5 Resource Consumption Profiles

Metric	Median	P90	P99	Implication
max_cpu / avg_cpu (burst)	6.3×	23.3×	46.0×	Peak memory prediction essential; avg understates demand
max_mem / avg_mem (burst)	1.00×	1.03×	1.66×	Memory is near-stable; avg is a reliable signal
req_memory vs avg_mem corr.	—	—	r = 0.52	Moderate: user requests have predictive value but need correction
req_cpu vs max_cpu corr.	—	—	r = 0.069	Negligible: declared CPU request cannot be used as a proxy
Memory overcommit ratio	1.64×	18.3×	100×	Confirms need for probabilistic safety margin (Cantelli cone)

CPU exhibits extreme burst behaviour (median 6.3×, max >600,000×). Memory is comparatively stable (median burst ratio 1.00×). Zero-CPU rows account for 1.17% of measurements; these correspond to pre-execution binary-copy periods documented in the trace specification.

References

Wilkes, J., Reiss, C., Deng, N., Haque, M. E., & Tirmazi, M. (2020). Google cluster-usage traces v3 [Dataset documentation]. Google. CC-BY licence.

Tirmazi, M., Barker, A., Deng, N., Haque, M. E., Qin, Z. G., Hand, S., Harchol-Balter, M., & Wilkes, J. (2020). Borg: the next generation. Proceedings of EuroSys 2020.

Verma, A., Pedrosa, L., Korupolu, M. R., Oppenheimer, D., Tune, E., & Wilkes, J. (2015). Large-scale cluster management at Google with Borg. Proceedings of EuroSys 2015.

Group sources

Data Extraction Folder:

https://guscanada-my.sharepoint.com/personal/tha_hmu7569_myunfc_ca/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Ftha%5Fhmu7569%5Fmyunfc%5Fca%2FDocuments%2F1%2ECapstone%2F3%2EPredictive%20Layer%2F1%2EData%20Extraction%2FTenant%5Fdataset&viewid=7bc35ee7%2Db01e%2D430f%2D92d0%2D8b6f03ab468e

Tenant Selection Methodology:

[tenant_selection_report \(2\).pdf](#)

EDA Folder:

https://guscanada-my.sharepoint.com/personal/tha_hmu7569_myunfc_ca/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Ftha%5Fhmu7569%5Fmyunfc%5Fca%2FDocuments%2F1%2ECapstone%2F3%2EPredictive%20Layer%2F2%2EEDA%2FTenant%20dataset&viewid=7bc35ee7%2Db01e%2D430f%2D92d0%2D8b6f03ab468e

Database csv:

[tenant_full_v1.csv](#)